22 INTRODUCTION TO ORGANIC CHEMISTRY

There are millions of different organic compounds. They all contain carbon and hydrogen, and often other elements such as oxygen, nitrogen and chlorine. Carbon atoms can join together to form chains and rings, which is why there are so many carbon compounds. This chapter introduces you to some of the important ideas that you need to know before you can start to understand organic chemistry.





▲ Figure 22.2 What do caffeine, aspirin and dyes have in common? They are all organic compounds.

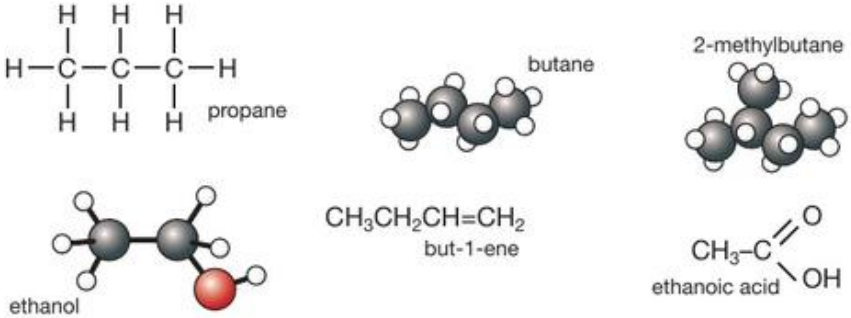
LEARNING OBJECTIVES

- Know that a hydrocarbon is a compound made up of hydrogen and carbon only
- Understand how to represent organic molecules using empirical formulae, molecular formulae, general formulae, structural formulae and displayed formulae
- Know what is meant by the terms homologous series, functional group and isomerism
- Understand how to name compounds (containing up to six carbon atoms) using the rules of International Union of Pure and Applied Chemistry (IUPAC) nomenclature
- Understand how to write the possible structural and displayed formulae of an organic molecule given its molecular formula
- Understand how to classify reactions of organic compounds as substitution, addition and combustion

In the next few chapters we will be looking at organic compounds. The term 'organic' was originally used because it was believed that organic compounds could only come from living things. Now it is used for any carbon compound except for the very simplest (carbon dioxide, carbon monoxide, the carbonates and the hydrogencarbonates).

Organic compounds can exist as chains, branched chains or rings of carbon atoms with hydrogens attached.

When you start doing organic chemistry, you are suddenly faced with a whole lot of new compounds with strange names and unfamiliar ways of drawing them. It can be quite scary!



▲ Figure 22.3 Organic chemistry involves a lot of new compounds.

HINT

Your school may have models you can use. Otherwise, you can make your own out of modelling clay and matchsticks or small nails to use as bonds.

The secret at the beginning is to spend a lot of time exploring the subject – understanding names, drawing structures and making models.

HYDROCARBONS

The simplest organic compounds are **hydrocarbons**. *These are molecules that contain carbon and hydrogen only*.

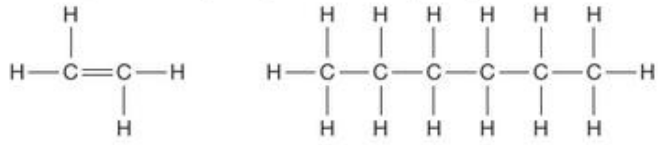
HINT

The word *only* is important here. Virtually all organic compounds contain carbon and hydrogen but hydrocarbons contain *only* carbon and hydrogen.

The carbon atoms are joined together with single, double or triple bonds. Carbon atoms are joined to hydrogen atoms by single bonds.

KEY POINT

Organic compounds can be described as **saturated** (containing only single C–C bonds) or **unsaturated** (containing double or triple C–C bonds). This is discussed in Chapters 24 and 25.



ethene hexane

▲ Figure 22.4 Examples of hydrocarbons

TYPES OF FORMULA FOR ORGANIC MOLECULES

EMPIRICAL FORMULAE

REMINDER

Empirical and molecular formula are introduced in Chapter 5.

An **empirical formula** tells you *the simplest whole number ratio of the atoms in a compound*. It can be calculated from experimental data. Without more information, it is not possible to identify the 'true' or 'molecular' formula. Sometimes the empirical formula is the same as the molecular formula. For example, the empirical formula of methane is CH_4, which is the same as its molecular formula. However, the empirical formula of ethane is CH_3 whereas its molecular formula is C_2H_6.

MOLECULAR FORMULAE

A **molecular formula** *counts the actual number of each type of atom present in a molecule*. For example, the molecular formula of butane is C_4H_{10} and the molecular formula of ethene is C_2H_4.

The molecular formula is a multiple of the empirical formula. For example, if a compound has empirical formula CH_2, the molecular formula could be C_2H_4, C_3H_6, C_4H_8, etc. The ratio of C:H must always be the same as in the empirical formula.

The molecular formula tells you nothing about the way the atoms are joined together. Both the compounds in Figure 22.5 have the same molecular formula.



▲ Figure 22.5 These two compounds have the same molecular formula but different structures.

Molecular formulae are used very rarely in organic chemistry because they don't give any useful information about the bonding in the molecule. You might use them in equations for the combustion of simple hydrocarbons, where the structure of the molecule doesn't matter. For example:

$$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$$

In almost all other cases, you use a structural or a displayed formula.

GENERAL FORMULAE

There are many different families of organic compounds, known as **homologous series**. Examples of homologous series are alkanes, alkenes, alcohols, carboxylic acids and esters. These homologous series will be discussed in more detail in later chapters.

A homologous series is a *series of compounds with similar chemical properties because they have the same functional group. Each member differs from the next by one –CH_2–.*

Members of a homologous series can be represented using a **general formula**. The first few members of the alkane series are shown in Table 22.1.

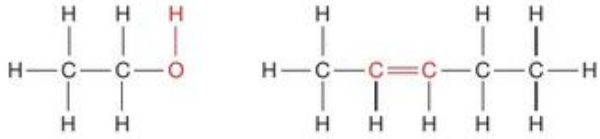
Table 22.1 The first few members of the alkane, alkene and alcohol homologous series

Alkanes	Alkenes	Alcohols
CH_4		CH_4O
C_2H_6	C_2H_4	C_2H_6O
C_3H_8	C_3H_6	C_3H_8O
C_4H_{10}	C_4H_8	$C_4H_{10}O$

In the case of alkanes, if there are n carbons in a molecule, there are always $2n + 2$ hydrogens. The general formula of alkanes is C_nH_{2n+2}. For methane, CH_4, there is 1 carbon atom in the molecule and $2 \times 1 + 2 = 4$ hydrogen atoms. For propane C_3H_8, there are 3 carbon atoms in the molecule and $2 \times 3 + 2 = 8$ hydrogen atoms. We can therefore work out that dodecane, which has 12 carbon atoms, will have 26 hydrogen atoms and the molecular formula $C_{12}H_{26}$.

Different homologous series usually have different general formulae. For example, the general formula for alkenes is C_nH_{2n} and that for alcohols is $C_nH_{2n+2}O$.

A **functional group** is *an atom or a group of atoms that determine the chemical properties of a compound*. All compounds in the same homologous series have the same functional group. For example, the functional group for alcohols is the –OH group and that for alkenes is C=C.



▲ Figure 22.6 Alcohols contain the –OH functional group and alkenes contain the C=C functional group.

HINT

More features of a homologous series are discussed on page 278.

HINT

A way of thinking about this is: the functional group is what makes the behaviour of a particular compound different from that of an alkane.

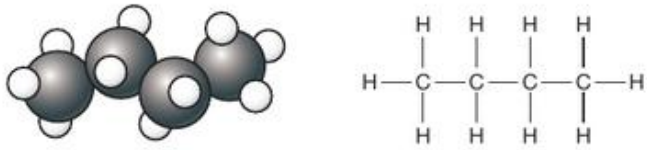
STRUCTURAL FORMULAE

A **structural formula** shows *how the atoms in a molecule are joined together*. There are two ways of representing structural formulae: they can be drawn as a **displayed formula** (full structural formula), or they can be written out in condensed form (*condensed structural formula*) by omitting all the carbon–carbon single bonds and carbon–hydrogen single bonds, for example $CH_3CH_2CH_3$. You need to be confident about using either way.

DISPLAYED FORMULAE

A displayed formula (sometimes called a full structural formula) shows *all* the bonds in the molecule as individual lines. You need to remember that each line represents *a pair of shared electrons in a covalent bond*.

Figure 22.7 shows a model of butane, together with its displayed formula. Notice that the way the displayed formula is drawn is different from the shape of the actual molecule. Displayed formulae are always drawn with the molecule straightened out and flattened. They do, however, show exactly how all the atoms are joined together.



▲ Figure 22.7 Butane. The angles between neighbouring C–H bonds in butane are shown as 90° in the two-dimensional displayed formula. In reality they are about 109.5° in a tetrahedral arrangement in three dimensions, similar to the arrangement of C–C bonds in a diamond structure (see Chapter 8, page 93).

REMINDER

It is important to remember when you draw structures that C always forms four bonds and H always forms one bond. Oxygen will form two bonds and a halogen (chlorine, bromine, etc.) atom one. If you are not sure about why these atoms form this number of bonds, you need to look back at Chapter 8.

KEY POINT

If you are asked to draw the structure for a molecule in an International GCSE exam, always draw it as a displayed formula. You can't lose any marks by doing this, whereas you might if you use the simplified form. If, on the other hand, you are simply writing a structure in an equation, for example, you can use whichever version you prefer.

HOW TO DRAW A STRUCTURAL FORMULA

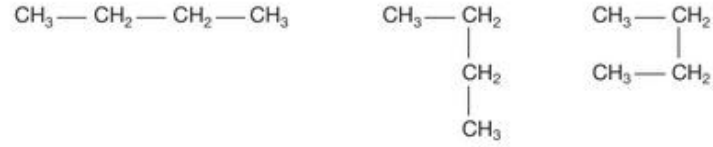
For anything other than the smallest molecules, drawing a fully displayed formula is very time-consuming. You can simplify the formula by writing, for example, CH_3 or CH_2 instead of showing all the carbon–hydrogen bonds.

The structural formula for butane can be shown as

$CH_3CH_2CH_2CH_3$ or $CH_3—CH_2—CH_2—CH_3$

These both show exactly how the atoms in the molecule are joined together. The corresponding molecular formula, C_4H_{10}, doesn't give you the same sort of useful information about the molecule.

All the structures in Figure 22.8 represent butane: even though they look different they are exactly the same molecule.



▲ Figure 22.8 All three structures represent butane. The convention is to write the structure with all the carbon atoms in a straight line.

HINT

The best way to understand that these are all the same is to make some models. If you don't have access to molecular models, use blobs of modelling clay joined together with pieces of matchstick or small nails to represent the bonds. You will find that you can change the shape of the model by rotating around the bonds. That's what happens in real molecules.

Each structure shows four carbon atoms joined together in a chain, but the chain has simply twisted. This happens in real molecules as well: the atoms can rotate around single carbon–carbon bonds, as shown in Figure 22.9.

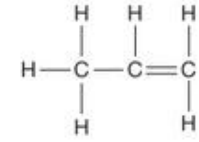


▲ Figure 22.9 These diagrams show the shape of a butane molecule better, but you would not draw these in an exam.

KEY POINT

Structural formulae may be drawn in various ways. Sometimes no bonds are shown and other times some bonds are shown. The easiest way to think about it is like this: if all the bonds are shown, then it is a displayed formula but if not all of them are shown (but you can still see what the structure is), then it is a structural formula.

A molecule like propene, C_3H_6, has a carbon–carbon double bond. This is shown by drawing two lines between the carbon atoms to show *the two pairs of shared electrons*. You would normally write this in a simplified structural formula as $CH_3CH{=}CH_2$.



▲ Figure 22.10 The displayed formula for propene

KEY POINT

In more advanced work the structural formula of propene would just be written as CH_3CHCH_2 but at International GCSE the C=C will normally be shown.

NAMING ORGANIC COMPOUNDS

Names for organic compounds can appear quite complicated, but they are simply a code that describes the molecule. Each part of a name tells you something specific about the molecule. One part of a name tells you how many carbon atoms there are in the longest chain, another part tells you whether there are any carbon–carbon double bonds, and so on.

CODING THE CHAIN LENGTH

Look for the code letters in the name – these are given in Table 22.2.

HINT

You have to learn these! The first four are the difficult ones because there isn't any pattern. However, a mnemonic can help, something like **M**onkeys **E**at **P**ink **B**ananas. Or you could think of one of your own that you can remember!
'**Pent**' means five (as in **pent**agon) and '**hex**' means six (as in **hex**agon).

Table 22.2 Coding the chain length

Code letters	Number of carbons in chain
meth	1
eth	2
prop	3
but	4
pent	5
hex	6

For example, **but**ane has a chain of four carbon atoms. **Prop**ane has a chain of three carbon atoms.

CODING FOR THE TYPE OF COMPOUND

ALKANES

Alkanes are *a homologous series of similar hydrocarbons (compounds of carbon and hydrogen only) in which all the carbons are joined to each other with single covalent bonds*. Compounds like this are coded with the ending '**ane**'. For example, eth*ane* is a two-carbon chain (because of 'eth') with a carbon–carbon single bond, $CH_3—CH_3$.

HINT

In more complicated molecules, the presence of the code 'an' in the name again shows that the carbons are joined by single bonds. For example, you can tell that propan-1-ol contains three carbon atoms ('prop') joined together by carbon–carbon single bonds ('an'). The coding on the end gives you more information about the molecule. You will meet this later in the chapter.

ALKENES

Alkenes are *a homologous series of hydrocarbons which contain a carbon–carbon double bond*. This is shown in their name by the ending '**ene**'. For example, eth*ene* is a two-carbon chain containing a carbon–carbon double bond, $CH_2=CH_2$. With longer chains, the position of the double bond could vary in the chain. This is shown by numbering the chain and noting which carbon atom the double bond *starts* from.

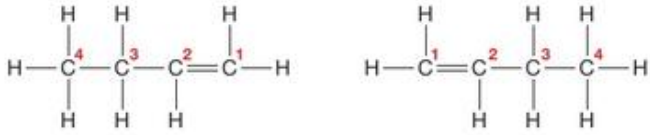
Table 22.3 Indicating the position of the double bond in the name of alkenes

Formula	Name	Description
$CH_2=CHCH_2CH_3$	but-1-ene	a four-carbon chain with a double bond starting on the first carbon
$CH_3CH=CHCH_3$	but-2-ene	a four-carbon chain with a double bond starting on the second carbon

HINT

Both parts of Figure 22.11 show the same molecule, but in one case it has been turned over so that what was originally on the left is now on the right, and vice versa. It would be silly to change the name every time the molecule moved! Both of the forms in Figure 22.11 are called but-1-ene.

How do you know which end of the chain to number from? The rule is that *you number from the end which produces the smaller numbers in the name*.



▲ Figure 22.11 Both structures represent but-1-ene.

CODING FOR BRANCHED CHAINS

Hydrocarbon chains can have side branches on them. You are only likely to come across two small side chains, shown in Table 22.4.

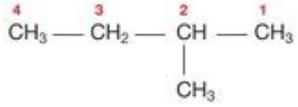
Table 22.4 Coding for branched chains

Side chain	Coded
CH_3–	methyl
CH_3CH_2–	ethyl

KEY POINT

Notice that the count of the number of carbons in the side chain is coded exactly as before: '**meth**' shows a *one*-carbon side chain, '**eth**' shows *two* carbons. The 'yl' shows that the group is attached to something else.

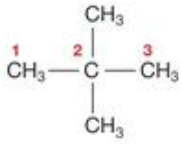
The name of a molecule is always based on the longest chain you can find in it. *The position of the side chain is shown by numbering the carbon atoms from the end of the longest chain which produces the smaller numbers in the name*, exactly as before with the carbon–carbon double bonds in alkenes.



▲ Figure 22.12 Number from the end that produces the smaller number.

The longest chain in the molecule in Figure 22.12 has four carbon atoms ('but') with no double bonds ('ane'). The name is based on butane. There is a methyl group branching from the number 2 carbon. (Remember to number from the end that produces the smaller number.) The compound is called 2-methylbutane.

Where there is more than one side chain, you describe the position of each of them.



▲ Figure 22.13 Describe the position of each side chain.

The longest chain in the molecule in Figure 22.13 has three carbon atoms and no double bonds. Therefore the name is based on propane.

There are two methyl groups attached to the second carbon. The compound is 2,2-dimethylpropane. The '*di*' in the name shows the presence of the two methyl groups. '2,2' shows that they are both on the second carbon atom.

You can reverse this process and draw a structural formula from a name. All you have to do is decode the name.

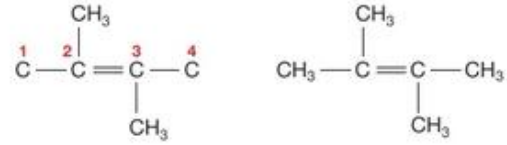
For example, what is the structural formula for 2,3-dimethylbut-2-ene?

Start by looking for the code for the longest chain length. 'but' shows a four-carbon chain. 'ene' shows that it contains a carbon–carbon double bond starting on the second carbon atom ('-2-ene'). There are two methyl groups ('dimethyl') attached to the second and third carbon atoms in the chain ('2,3-'). All you have to do now is to fit all this together into a structure.

HINT

It is much more important to be able to decode names to give structures than the other way around. If you don't know what a teacher or an examiner is talking about, you are completely lost!

Start by drawing the structure without any hydrogen atoms on the main chain. It doesn't matter whether you draw the CH_3 groups pointing up or down. Then add enough hydrogens so that each carbon atom forms four bonds. Figure 22.14 shows how your thinking would work.



▲ Figure 22.14 First draw the carbon structure, then add hydrogens.

KEY POINT

Warning! Don't confuse the word isomer with isotope. *Isotopes* are different atoms of the same element with the same atomic number but different mass numbers.

STRUCTURAL ISOMERISM

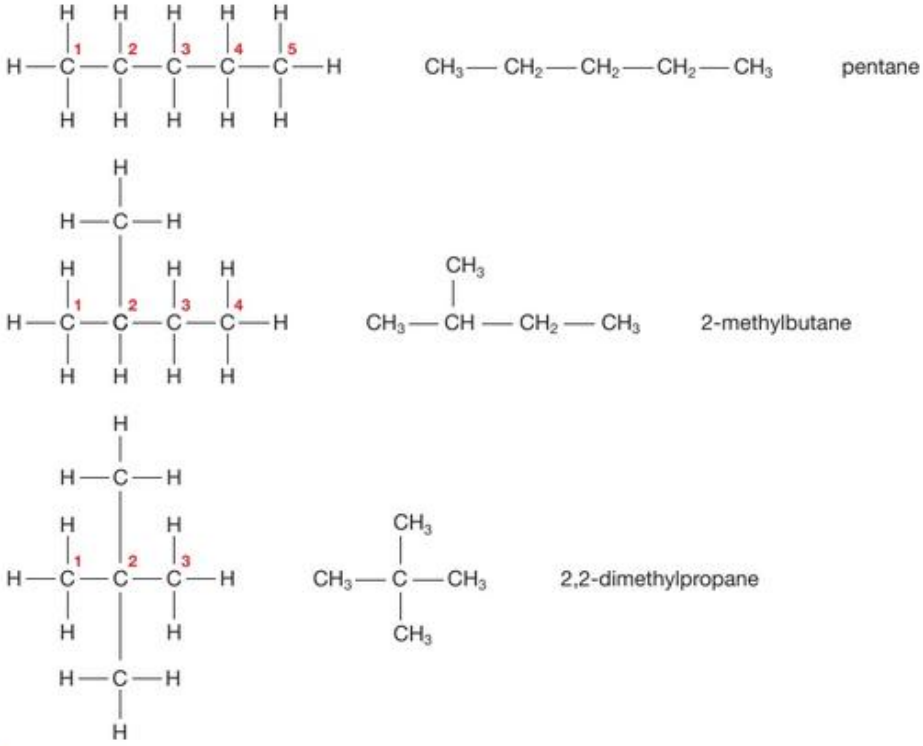
Structural isomers *are molecules with the same molecular formula, but different structural formulae.*

Examples will make this clear.

STRUCTURAL ISOMERISM IN THE ALKANES

ISOMERS OF C_5H_{12}

If you had some molecular models and picked out five carbon atoms and 12 hydrogen atoms, you would find it was possible to join them together in more than one way. The different molecules formed are known as isomers. All have the molecular formula C_5H_{12}, but different structures.



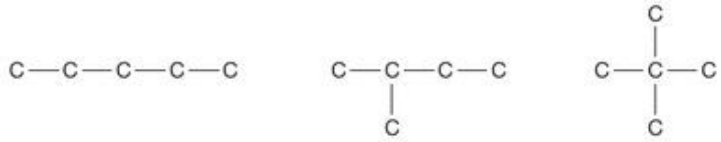
▲ Figure 22.15 There are three isomers for C_5H_{12}. These can be shown as either displayed formulae or structural formulae.

If you look carefully at the structures in Figure 22.15 you can see that you couldn't change one into the other simply by bending or twisting the molecule. You would have to break some bonds and reconnect the atoms. That's a simple way of telling that you have isomers.

HINT

It is sometimes easier to see that the molecules are different by drawing only the C atoms joined together, without the hydrogens.

But you must show the hydrogen atoms in the exam!



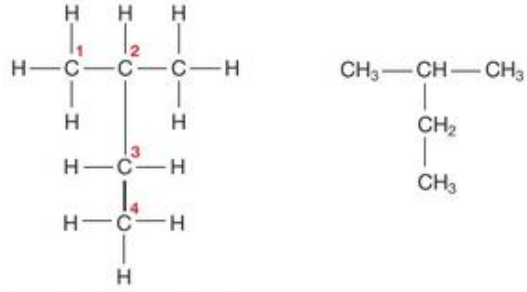
REMINDER

A straight chain is an unbranched chain.

The **straight chain** isomer is called pentane. The *branched chain* isomer in the middle, which has a four-carbon chain ('butane') and a methyl group on the second carbon, is called 2-methylbutane. The final *branched chain* isomer, which has a three-carbon chain ('propane') and two methyl groups on the second carbon, is called 2,2-dimethylpropane. The prefix '*di*' indicates the presence of two identical branches in the molecule.

Students frequently think they can find another isomer as well. If you look closely at this 'fourth' isomer (Figure 22.16) you will see that it is just 2-methylbutane rotated in space.

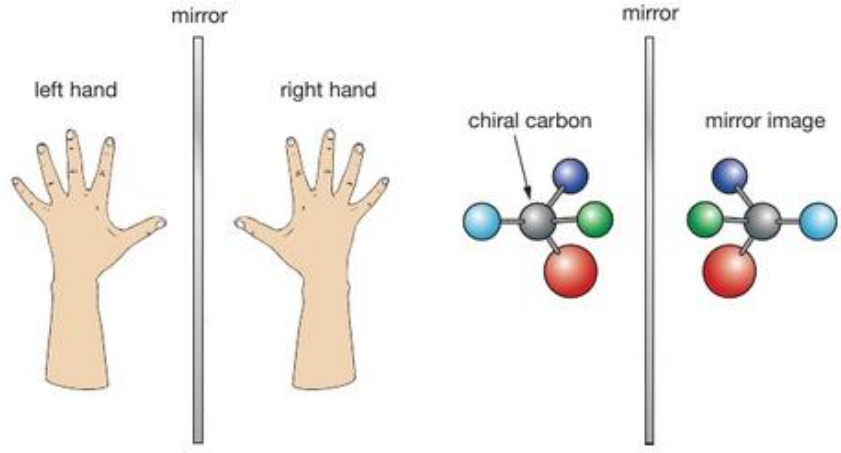
To avoid this sort of problem, always draw your isomers so that *the longest carbon chain is drawn horizontally* (or draw them out again without the H atoms).



▲ Figure 22.16 There is no fourth isomer for C_5H_{12}.

LOOKING AHEAD

To be precise, the type of isomerism discussed here should be called structural isomerism: molecules have the same molecular formula but different structural formulae. The atoms in the different isomers are joined in a different order. There are also other forms of isomerism and one important type is called optical isomerism. Optical isomers have atoms joined in the same order (i.e. they have the same structural formula), but are arranged differently in three-dimensional space. These isomers are mirror images of each other but cannot be superimposed. A good analogy is to look at your left and right hands. They are exactly the opposite of each other but cannot lie on top of another and align perfectly (Figure 22.17).



▲ Figure 22.17 Optical isomers are mirror images of each other and cannot be superimposed.

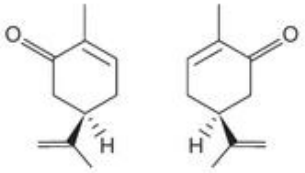
Optical isomerism occurs when a molecule has a *chiral centre*, that is to say it has a carbon atom that is connected to four different atoms or groups of atoms. These molecules have a tetrahedral shape around the chiral centre (molecular shape was mentioned in Chapter 8, pages 88–89). Figure 22.17 shows two molecules that are mirror images of each other. However, if you rotate one of the molecules 180° in space, it cannot be put on top of the other.

Optical isomers have identical chemical properties, except when they interact with something that is also chiral. For example, apart from glycine, all naturally occurring amino acids have optical isomers, which means that all the proteins in our body are chiral. This is why we can distinguish the two optical isomers in

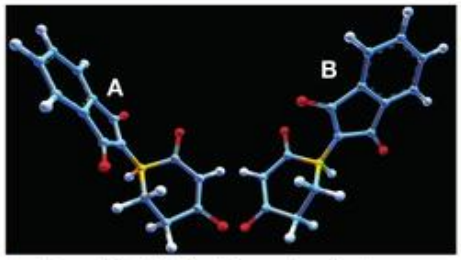
Figure 22.18. One smells of spearmint and the other of caraway seed because the two molecules interact differently with the protein receptors in our noses.

EXTENSION WORK

The optical isomers are assigned a prefix (*R* or *S*) to show the configuration of atoms around the chiral centre.



▲ Figure 22.18 Two optical isomers of carvone



▲ Figure 22.19 Optical isomerism in drug molecules

An important application of optical isomerism is in the synthesis of pharmaceutical drugs. You might have heard of the thalidomide tragedy. Back in the 1960s in Germany, the molecule B shown in Figure 22.19 was found to have anti-nausea effects and was therefore used to treat morning sickness in pregnant woman. Unfortunately, molecule B was being manufactured together with its optical isomer A and little was known about the molecule A until many children born in that era were found to have deformed limbs. These days, many drug companies devote a great deal of money to separating optical isomers of new drugs just in case they have side effects.

STRUCTURAL ISOMERISM IN THE ALKENES

ETHENE AND PROPENE

Ethene, $CH_2{=}CH_2$, doesn't have any isomers. Propene, $CH_3CH{=}CH_2$, doesn't have a structural isomer that is still an alkene.

EXTENSION WORK

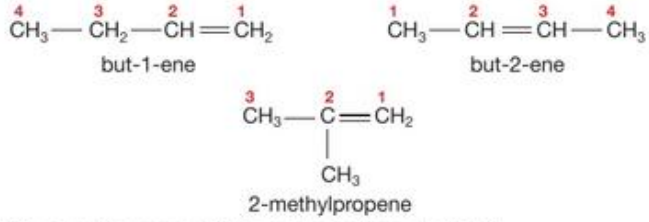
There is a structural isomer of C_3H_6 that doesn't have a carbon–carbon double bond. It has three carbon atoms joined in a ring. It is called cyclopropane, which indicates 'a ring of three carbons with only single bonds between them'.

ISOMERS OF C_4H_8

There are three structural isomers with the molecular formula C_4H_8 and containing a carbon–carbon double bond.

HINT

If you aren't comfortable with the names, see pages 258–260 on organic names.



▲ Figure 22.20 Structural isomers with the molecular formula of C_4H_8.

Notice the way that you can vary the position of the double bond as well as branching the chain.

CHEMISTRY ONLY

A QUICK INTRODUCTORY LOOK AT THE ALCOHOLS

Alcohols are *a homologous series of compounds which all contain an –OH functional group attached to a hydrocarbon chain*. This is coded for in the name by the ending '**-ol**'.